

# Seonghwan Seo

PhD .Candidate | Korea Advanced Institute of Science and Technology

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## RESEARCH INTERESTS

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Generative Modeling

Generative Flow Networks (GFlowNets); diffusion and flow-based models

AI for Drug Discovery

Synthesis-aware molecular design; structure-based drug discovery

Biomolecular Design

Protein design; protein folding and biomolecular co-folding

## EXPERIENCE

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Intelligent Chemistry Lab, KAIST

Daejeon, South Korea

AI for Drug Discovery

June 2019 - Present

Supervisor: Woo Youn Kim

- Development of an unbiased drug-likeness scoring model
- Development of a deep learning framework for fully automated receptor-based pharmacophore modeling
- Research on synthesis-aware drug discovery
- Development of trillion-scale chemical library search technology with  $\Phi$ Molecules & HITS
- National project on K-Fold, a next-generation foundation model for biomolecular co-folding

HITS Inc.

Seoul, South Korea

AI Research Intern

Dec. 2020 - Aug. 2022

Manager: Jaechang Lim

- Development of a reaction template-based generative model for synthesizable molecular design with desired properties
- Development of a transformer-based model for synthesis pathway generation

## EDUCATION

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Korea Advanced Institute of Science and Technology (KAIST)

Daejeon, South Korea

PhD .student in Chemistry

Aug. 2022 - Feb. 2028 (expected)

Supervisor: Woo Youn Kim

Korea Advanced Institute of Science and Technology (KAIST)

Daejeon, South Korea

B.S. in Chemistry & Computer Science (Double Major)

Feb. 2018 - Aug. 2022

Supervisor: Youngmin Rhee

GPA : 3.99/4.3

## PUBLICATIONS

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First author

\* indicates equal contribution; † indicates corresponding author

- Boltz2ES1: Accurate Enzyme Specificity Prediction with Co-folding Foundation Model [\[link\]](#)  
Xivei Cheng\*, Seonghwan Seo\*, Jihang Chen, Songlin Jiang, Pengkang Guo, Wengong Jin†  
ICLR 2026 Workshop on Generative and Experimental Perspectives for Biomolecular Design (GEM)
- Compositional Flows for 3D Molecule and Synthesis Pathway Co-design [\[link\]](#)  
Tony Shen\*, Seonghwan Seo\*, Ross Irwin, Kieran Didi, Simon Olsson, Woo Youn Kim, Martin Ester†  
International Conference on Machine Learning (ICML), 2025
- Generative Flows on Synthetic Pathway for Drug Design [\[link\]](#)  
Seonghwan Seo, Minsoo Kim, Tony Shen, Martin Ester, Jinkyoo Park, Sungsoo Ahn, Woo Youn Kim†  
International Conference on Learning Representations (ICLR), 2025
- PharmacoNet: deep learning-guided pharmacophore modeling for ultra-large-scale virtual screening [\[link\]](#)  
Seonghwan Seo, Woo Youn Kim†  
Chemical Science, 2024
- Molecular Generative Model via Retrosynthetically Prepared Chemical Building Block Assembly [\[link\]](#) [\[cover\]](#)  
Seonghwan Seo, Jaechang Lim & dagger, Woo Youn Kim†  
Advanced Science, 2023 (selected for a journal cover)

- Drug-likeness scoring based on unsupervised learning [\[link\]](#)  
Kyunghoon Lee\*, Jinho Jang\*, Seonghwan Seo\*, Jaechang Lim & dagger, Woo Youn Kim†  
Chemical Science, 2022

#### Co-author

- Atom-level Protein Representation Learning Improves Protein Structure Prediction [\[link\]](#)  
Taewon Kim, Hyosoon Jang, Hyunjin Seo, Seonghwan Seo, Hyeonwoo Kim, Wonho Chung, Mingyeong Shin, Wooyoung Kim, Sungsoo Ahn†  
arXiv preprint, 2026
- Generalized Multi-State Protein Design with AlphaFold3 [\[link\]](#)  
Xivei Cheng, Pengkang Guo, Seonghwan Seo, Casper A. Goverde, Wengong Jin†  
ICLR 2026 Workshop on Generative and Experimental Perspectives for Biomolecular Design (GEM)
- TacoGFN: Target Conditioned GFlowNet for Structure-based Drug Discovery [\[link\]](#)  
Tony Shen, Seonghwan Seo, Grayson Lee, Mohit Pandey, Jason R Smith, Artem Cherkasov, Woo Youn Kim, Martin Ester†  
Transactions on Machine Learning Research (TMLR), 2024

## ACADEMIC SERVICE

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- Organizer of the Generative AI and Biology (GenBio) at ICM L 2026 in Seoul
- Reviewer (Conferences): ICLR (2025-2026), NeurIPS (2026)
- Reviewer (Journals): Nature Communications (2024)

## PRESENTATIONS

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#### Invited

- "Molecular Generative Model via Retrosynthetically Prepared Chemical Building Block Assembly"  
Oral workshop and tutorial at AIBIO, Artificial Intelligence Institute, Seoul National University, Seoul, South Korea (Nov. 2022)

#### Contributed

- "Compositional Flows for 3D Molecule and Synthesis Pathway Co-design"  
Oral spotlight presentation at ICLR 2025 Workshop on AI for Accelerated Materials Discovery (AIMat), Singapore. (Apr. 2025)
- "Compositional Flows for 3D Molecule and Synthesis Pathway Co-design"  
Oral spotlight presentation at ICLR 2025 Workshop on Generative and Experimental Perspectives for Biomolecular Design (GEM), Singapore. (Apr. 2025)
- "Generative Flows on Synthetic Pathway for Drug Design"  
Poster presentation at ICLR 2025, Singapore. (Apr. 2025)
- "Generative Flows on Synthetic Pathway for Drug Design"  
Poster presentation at NeurIPS 2024 Workshop on AI for New Drug Modalities (AIDrugX), Vancouver, Canada. (Dec. 2024)
- "PharmacoNet: Accelerating Large-Scale Virtual Screening by Deep Pharmacophore Modeling"  
Poster presentation at NeurIPS 2023 Workshop on New Frontiers of AI for Drug Discovery and Development (A4D3), New Orleans, LA, USA. (Dec. 2023)
- "Molecular Generative Model via Retrosynthetically Prepared Chemical Building Block Assembly"  
Poster presentation at 2023 Accelerate Conference, Toronto, Canada (Aug. 2023)

## AWARDS & SCHOLARSHIPS

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- TTA President Award  
Korea AI-Champion Competition, 2025
- KAIST Alumni Scholarship  
KAIST Alumni Association, 2021 - 2022
- Younghoon Lee Scholarship  
Department of Chemistry, KAIST, 2021
- Model Student Awards  
KAIST, 2021
- Dean's List  
College of Natural Science, KAIST, 2021

Government of South Korea, 2020 - 2021

- Dean's List  
School in Freshman, KAIST, 2019

## SKILLS

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Languages

Korean (native), English (upper-intermediate)

Deep Learning Tools

Python, PyTorch, PyTorch Geometric, PyTorch Lightning, Numba, Pandas

Chemistry Tools

RDKit, OpenBabel, Gemmi, Vina, Smiles, rDock